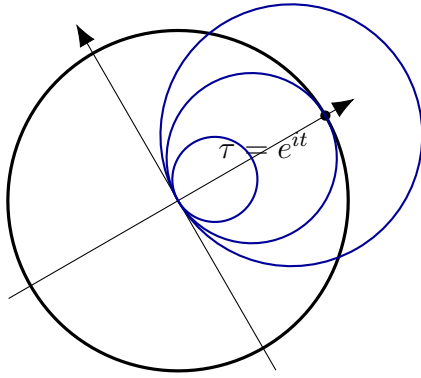


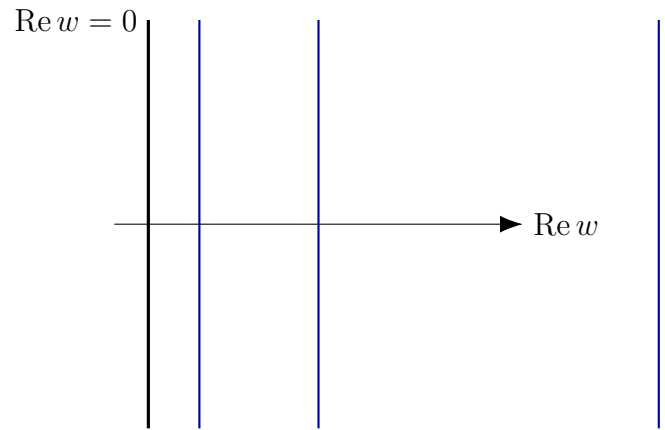
Two-by-two mapping charts for  $K_t(z) = \frac{e^{it} + z}{e^{it} - z}$

**Description.** The Möbius kernel  $K_t$  maps  $\mathbb{D}$  conformally onto  $\{w : \operatorname{Re} w > 0\}$ . In the  $z$ -plane, the level sets  $\{\operatorname{Re} K_t = c\}$  (blue) and  $\{\operatorname{Im} K_t = b\}$  (red) are two orthogonal pencils of circles, both tangent to  $\partial\mathbb{D}$  at  $\tau = e^{it}$ . They correspond, respectively, to vertical and horizontal lines in the image half-plane.

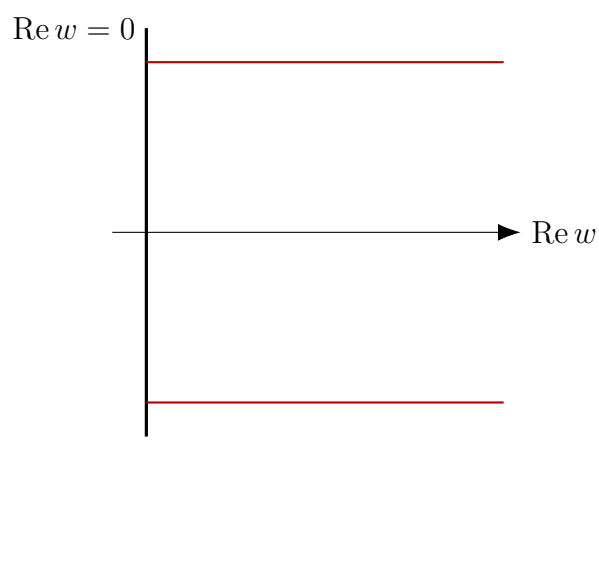
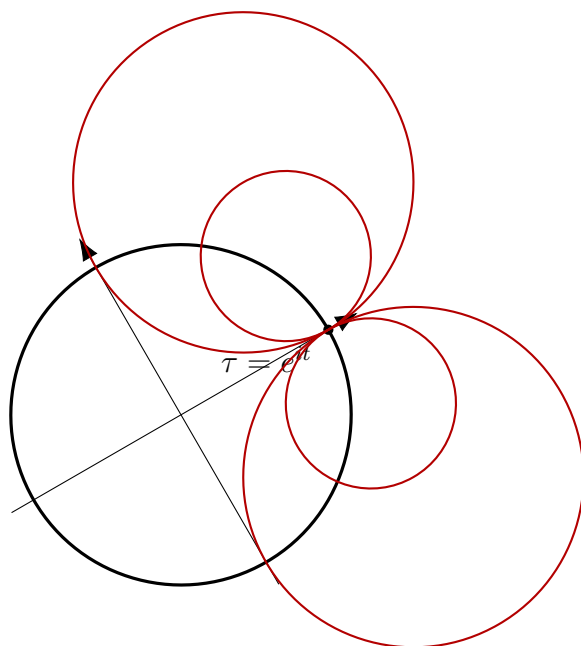
**Domain:  $z$ -plane (vertical family)**



**Image:  $w$ -plane (vertical lines)**



Blue circles  $\{\operatorname{Re} K_t = c\}$  in the domain  $\longleftrightarrow$  blue vertical lines  $\{\operatorname{Re} w = c\}$  in the image.

Domain:  $z$ -plane (horizontal family)Image:  $w$ -plane (horizontal lines)

Red circles  $\{\text{Im } K_t = b\}$  in the domain  $\longleftrightarrow$  red horizontal lines  $\{\text{Im } w = b\}$  in the image.

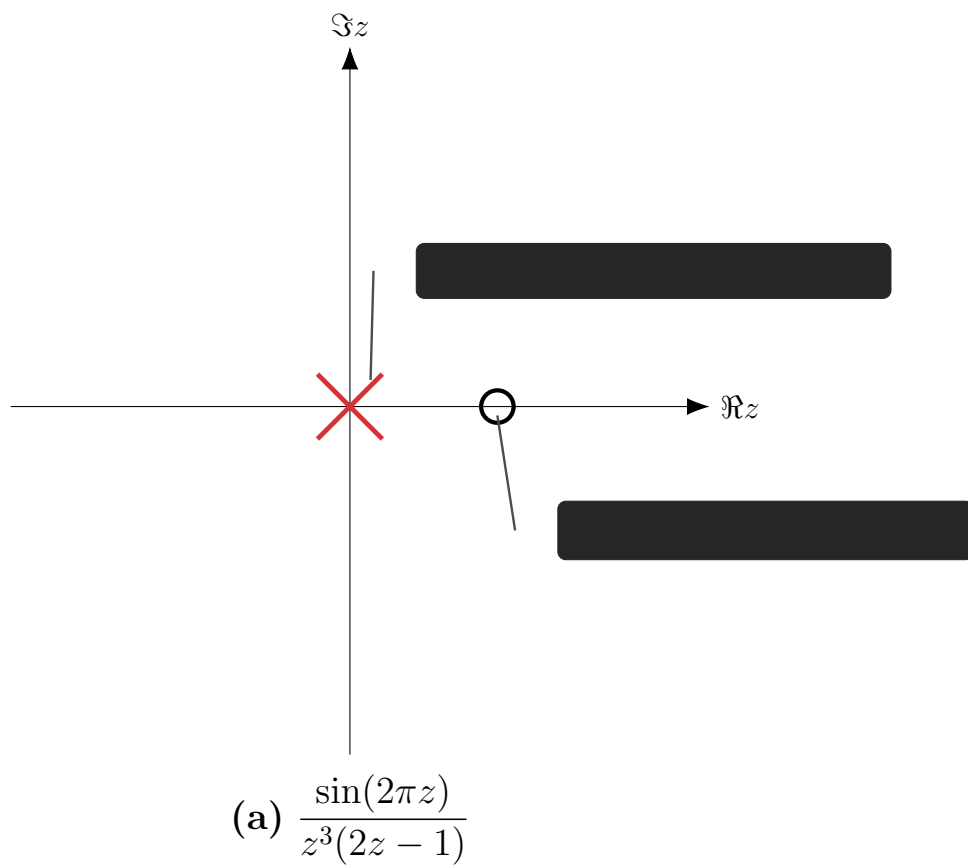


Figure 1: **Panel (a).** The function  $\sin(2\pi z)/(z^3(2z-1))$  has a double pole at  $z=0$  with residue  $-4\pi$  and a removable singularity at  $z=\frac{1}{2}$  (residue 0).

## Conformal maps: an overview catalogue (no details)

1. **Cayley (disk  $\leftrightarrow$  right half-plane).**

$$\boxed{w = \frac{1+z}{1-z}} \quad \text{inverse: } z = \frac{w-1}{w+1}.$$

2. **Half-plane  $\rightarrow$  half-plane (three-point normalization).**

$$\boxed{w = \frac{(z-a)(c-b)}{(z-b)(c-a)}}.$$

3. **Disk automorphisms.**

$$\boxed{\phi_{a,\theta}(z) = e^{i\theta} \frac{z-a}{1-\bar{a}z}}, \quad |a| < 1.$$

4. **Strip  $|\Im z| < h/2 \rightarrow$  right half-plane.**

$$\boxed{w = \exp\left(\frac{\pi z}{h}\right)}.$$

5. **Strip  $|\Im z| < h/2 \rightarrow$  unit disk.**

$$\boxed{\zeta = \frac{e^{\pi z/h} - 1}{e^{\pi z/h} + 1}}.$$

6. **Strip  $\rightarrow$  strip (height scaling).**

$$\boxed{z \mapsto \frac{h_2}{h_1} z}.$$

7. **Band  $a < \Im z < b \rightarrow$  centered strip.**

$$\boxed{z \mapsto \frac{\pi}{b-a} \left( z - i \frac{a+b}{2} \right)}.$$

8. **Sector  $\{0 < \arg z < \alpha\} \rightarrow$  upper half-plane.**

$$\boxed{w = z^{\pi/\alpha}} \quad (\text{principal branch}).$$

9. **Wedge  $\rightarrow$  disk.**

Compose item (8) with Cayley (item 1).

10. **Upper half-plane  $\rightarrow$  disk (prescribed triple).**

$$\boxed{\Phi(z) = \frac{(z-a)}{(z-\bar{a})} \cdot \frac{(b-\bar{a})}{(b-a)}}.$$

11. **Annulus**  $r < |z| < 1 \leftrightarrow$  **strip**.

$$\boxed{w = \log z} \quad \text{maps to} \quad \log r < \Re w < 0.$$

12. **Annulus**  $\rightarrow$  **disk**.

Use  $w = \log z$  then item (5).

13. **Half-plane minus disk**  $\rightarrow$  **annulus**.

$$\boxed{\zeta = \frac{z - \alpha}{z + \alpha}}, \quad \alpha^2 = a^2 - b^2.$$

14. **Disk with off-center hole**  $\rightarrow$  **annulus**.

Center via  $\phi_a$  (item 3), then items (11)–(12).

15. **Punctured plane**  $\leftrightarrow$  **cylinder**.

$$\boxed{w = \log z} \quad (\text{period } 2\pi i).$$

16. **Strip**  $\rightarrow$  **punctured plane**.

$$\boxed{z \mapsto e^z}.$$

17. **Ray-slit plane**  $\mathbb{C} \setminus [0, \infty) \rightarrow$  **half-plane**.

$$\boxed{w = \sqrt{z}} \quad (\text{choose branch}).$$

18. **Segment-slit plane**  $\mathbb{C} \setminus [a, b] \rightarrow$  **half-plane**.

$$\boxed{w = \sqrt{\frac{z - a}{z - b}}}.$$

19. **Exterior of**  $[-1, 1] \leftrightarrow$  **exterior unit circle (Joukowski)**.

$$\boxed{w = z + \frac{1}{z}}, \quad z = \frac{w \pm \sqrt{w^2 - 4}}{2}.$$

20. **Exterior of ellipse**  $\leftrightarrow$  **exterior unit circle**.

$$\boxed{w = \frac{1}{2} \left( cz + \frac{1}{cz} \right)} \quad (c \text{ from axes}).$$

21. **Right half-plane**  $\rightarrow$  **vertical strip**.

$$\boxed{z \mapsto \frac{1}{\pi} \log \frac{z - 1}{z + 1}}.$$

22. **Upper half-plane  $\rightarrow$  rectangle (SC/elliptic).**

$$\zeta = \int^z \frac{dt}{\sqrt{(t-a)(t-b)(t-c)(t-d)}}.$$

23. **Rectangle  $\rightarrow$  disk (Jacobi sn).**

$$\zeta = \operatorname{sn}\left(\frac{K(k)}{L} z; k\right).$$

24. **Half-disk  $\leftrightarrow$  half-plane.**

Restriction of Cayley (item 1) and inverse.

25. **Lens (intersection of two disks)  $\rightarrow$  disk.**

Möbius normalizes the circles; then a power map; then Cayley.

26. **Half-strip  $\{x > 0, 0 < \Im z < h\} \rightarrow$  half-disk.**

$$\exp(\pi z/h) \quad \text{then power, then Cayley.}$$

27. **Polygon  $\rightarrow$  half-plane (Schwarz–Christoffel).**

$$\Phi'(z) = C \prod_k (z - z_k)^{\alpha_k - 1}, \quad \sum \alpha_k = 2.$$

28. **Upper half-plane  $\rightarrow$  slit half-plane.**

$$w = z + \frac{1}{z} \quad \text{then a real Möbius adjustment.}$$

29. **Disk  $\rightarrow$  disk with boundary arc removed.**

Automorphism to put endpoints at  $\pm 1$ , then boundary power  $z^\lambda$ .

30. **Finite Blaschke product (disk  $\rightarrow$  disk, degree  $n$ ).**

$$B(z) = e^{i\theta} \prod_{k=1}^n \frac{z - a_k}{1 - \bar{a}_k z}.$$

## 1. Cayley transform: $\mathbb{D} \leftrightarrow$ right half-plane

Map and inverse.

$$w = \frac{1+z}{1-z} \quad \text{maps} \quad \mathbb{D} = \{|z| < 1\} \quad \text{onto} \quad \{\Re w > 0\}, \quad z = \frac{w-1}{w+1}$$

is the inverse.

**Why  $\Re w > 0$ .**

Compute

$$\Re\left(\frac{1+z}{1-z}\right) = \frac{1-|z|^2}{|1-z|^2}.$$

Thus  $\Re w > 0$  for  $|z| < 1$ , and  $\Re w = 0$  for  $|z| = 1$  (except  $z = 1$ , which maps to  $w = \infty$ ).  
Therefore

$$|z| < 1 \iff \Re w > 0, \quad |z| = 1 \iff \Re w = 0.$$

**Boundary correspondence.**

$z = 0 \mapsto w = 1$ ,  $z \rightarrow 1^- \mapsto w \rightarrow +\infty$ , and  $\partial\mathbb{D} \setminus \{1\}$  maps to the line  $\Re w = 0$ .

Circles/lines orthogonal to  $\partial\mathbb{D}$  map to *vertical* lines in the  $w$ -plane.

## Cayley transform: $\mathbb{D} \leftrightarrow$ right half-plane

Consider the Cayley transform

$$w = \phi(z) := \frac{1+z}{1-z}, \quad z = \phi^{-1}(w) = \frac{w-1}{w+1}.$$

Then

$$\Re w = \Re\left(\frac{1+z}{1-z}\right) = \frac{1-|z|^2}{|1-z|^2}.$$

Hence

$$|z| < 1 \iff \Re w > 0, \quad |z| = 1, z \neq 1 \iff \Re w = 0.$$

Thus the Cayley transform maps the unit disc  $\mathbb{D} = \{|z| < 1\}$  biholomorphically onto the right half-plane  $\{w : \Re w > 0\}$ , and the unit circle  $\partial\mathbb{D} \setminus \{1\}$  onto the imaginary axis.

### Images of lines and circles

Because  $\phi$  is a Möbius transformation, it maps generalized circles (lines or circles) in the  $z$ -plane to generalized circles in the  $w$ -plane. In particular:

- For  $0 < r < 1$ , the circle  $|z| = r$  is orthogonal to the unit circle and is mapped to a vertical line in the right half-plane. A direct computation gives

$$|z| = r \iff \Re w = \frac{1+r^2}{1-r^2} > 0.$$

- Radii  $z = te^{i\theta}$ ,  $0 \leq t < 1$ , become circular arcs in the right half-plane, orthogonal to the boundary line  $\Re w = 0$ .
- Curves tangent to the unit circle at  $z = 1$  are mapped to horizontal lines in the  $w$ -plane (parallel to the boundary).

### Mapping charts

Figure 2 shows a polar grid inside the unit disc: several concentric circles and radii. In Figure 3 we see its image under the Cayley transform: concentric circles become vertical lines, radii become circular arcs, and the unit circle maps to the imaginary axis.

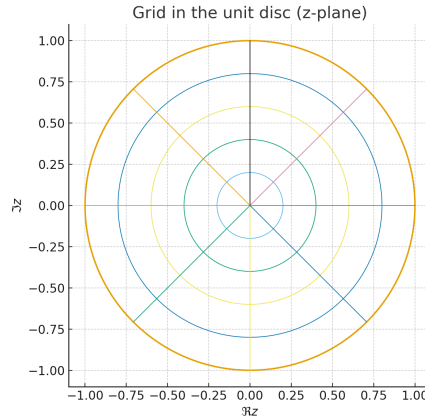


Figure 2: Grid in the unit disc ( $z$ -plane): unit circle, concentric circles, and radii.

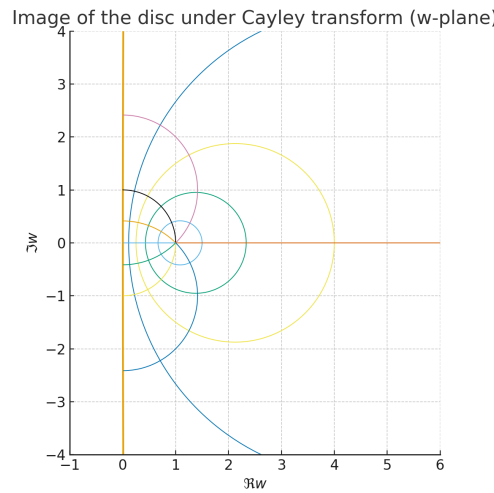


Figure 3: Image of the disc under the Cayley transform in the  $w$ -plane: the unit circle becomes the imaginary axis, concentric circles become vertical lines, and radii become circular arcs orthogonal to the boundary.

### Vertical and horizontal lines

To visualise more explicitly how vertical and horizontal lines in the right half-plane look in the disc, consider the inverse Cayley transform

$$z = \phi^{-1}(w) = \frac{w - 1}{w + 1}.$$

**Vertical lines.** Fix  $c > 0$  and consider the vertical line

$$L_c := \{ w = c + it : t \in \mathbb{R} \}.$$

Its preimage  $\phi^{-1}(L_c)$  is a circle inside the unit disc, orthogonal to  $\partial\mathbb{D}$ . In particular, every vertical line  $\Re w = c$  intersects the boundary  $\Re w = 0$  at right angles, and its preimage is a circle intersecting  $|z| = 1$  at right angles.

**Horizontal lines.** Fix  $d \in \mathbb{R}$  and consider the horizontal line

$$H_d := \{ w = s + id : s > 0 \}.$$

Its preimage  $\phi^{-1}(H_d)$  is a circle tangent to the unit circle at the point  $z = 1$ . Thus horizontal lines in the half-plane correspond to circles in the disc that “aim at” the boundary point mapped to infinity.

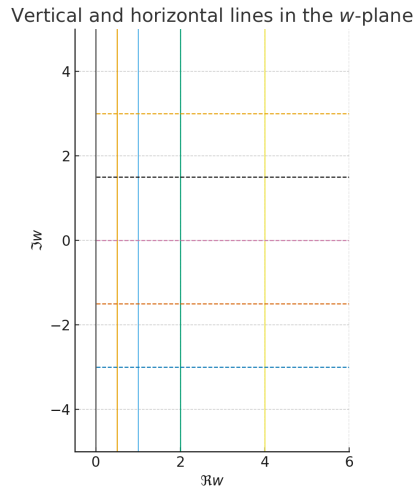


Figure 4: Families of vertical (solid) and horizontal (dashed) lines in the  $w$ -plane. The imaginary axis  $\Re w = 0$  is the boundary of the image of the disc.

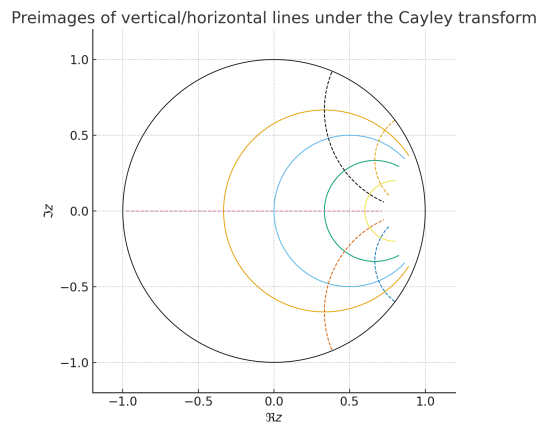


Figure 5: Preimages in the  $z$ -plane of the vertical and horizontal lines from Figure 4. Vertical lines become circles orthogonal to the unit circle, while horizontal lines become circles tangent to  $\partial\mathbb{D}$  at  $z = 1$ .

### Vertical and horizontal chords in the disc

Besides the polar grid, it is instructive to see what happens to straight chords of the unit disc. Consider the vertical and horizontal segments

$$\{z = a + iy : a^2 + y^2 < 1\}, \quad \{z = x + ib : x^2 + b^2 < 1\},$$

for fixed real numbers  $a, b$  with  $|a|, |b| < 1$ . These are vertical and horizontal lines in the  $z$ -plane, truncated to lie inside the disc.

Under the Cayley transform

$$w = \phi(z) = \frac{1+z}{1-z},$$

each such chord is mapped to a generalized circle in the  $w$ -plane. In particular, the real axis segment  $(-1, 1) \subset \mathbb{R}$  (the chord  $\Im z = 0$ ) is mapped onto the positive real axis in the half-plane. Other vertical and horizontal chords are mapped to curved arcs lying entirely in the region  $\Re w > 0$ .

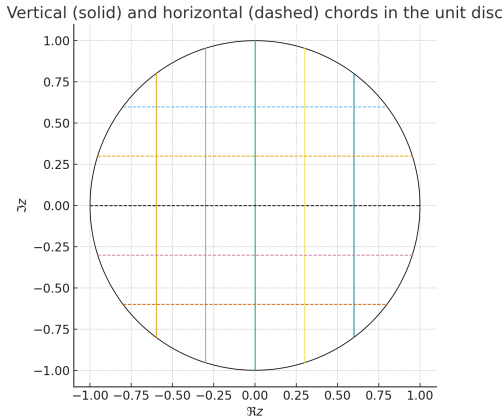


Figure 6: Vertical (solid) and horizontal (dashed) chords inside the unit disc.

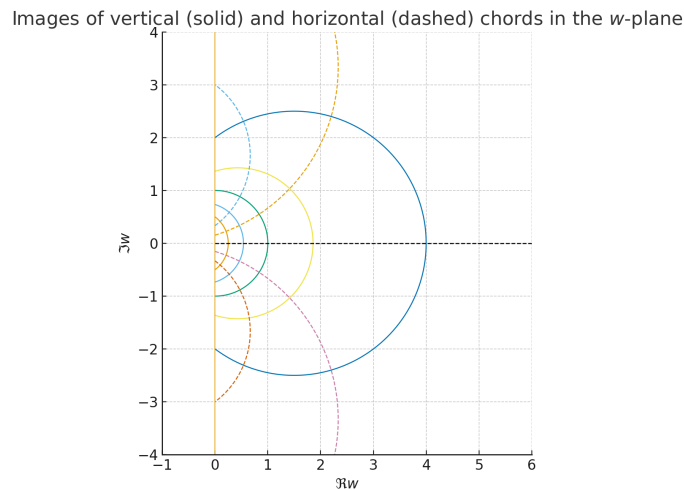


Figure 7: Images of the chords from Figure 6 under the Cayley transform. Each chord becomes a generalized circle lying in the right half-plane; the real axis segment maps to the positive real axis.

Fix real numbers  $a, b$  with  $|a|, |b| < 1$ . Consider the vertical and horizontal chords inside the unit disc

$$\{z = a + iy : a^2 + y^2 < 1\}, \quad \{z = x + ib : x^2 + b^2 < 1\}.$$

These are the intersections of the vertical line  $\Re z = a$  and the horizontal line  $\Im z = b$  with the unit disc  $|z| < 1$ .

## 2. Three-point Möbius normalization: half-plane $\leftrightarrow$ half-plane

**Cross-ratio map to the real axis.**

Let  $L$  be a line or circle, and choose three distinct points  $a, b, c \in L$ . Define

$$\Phi(z) = \frac{(z-a)(c-b)}{(z-b)(c-a)}.$$

Then  $\Phi(L) \subset \mathbb{R} \cup \{\infty\}$ , with

$$\Phi(a) = 0, \quad \Phi(b) = \infty, \quad \Phi(c) = 1,$$

and the two sides of  $L$  map to the half-planes  $\Im \Phi(z) \gtrless 0$ .

## Three-point Möbius normalization: half-plane $\leftrightarrow$ half-plane

**Cross-ratio map to the real axis**

Let  $L$  be a line or circle in the extended complex plane  $\widehat{\mathbb{C}}$ , and choose three distinct points  $a, b, c \in L$ . Define

$$\Phi(z) = \frac{(z-a)(c-b)}{(z-b)(c-a)}.$$

This is a Möbius transformation with

$$\Phi(a) = 0, \quad \Phi(b) = \infty, \quad \Phi(c) = 1.$$

Moreover, one has

$$\Phi(L) \subset \mathbb{R} \cup \{\infty\},$$

so  $L$  is mapped onto the real axis (possibly passing through  $\infty$ ), and the two sides of  $L$  are mapped to the two open half-planes  $\Im \Phi(z) > 0$  and  $\Im \Phi(z) < 0$ .

In other words, by fixing three marked points  $a, b, c$  on  $L$ , the cross-ratio map  $\Phi$  *normalizes* the configuration so that these points become  $0, \infty, 1$  and  $L$  itself becomes the real line.

**Example: a circle mapped to the real axis**

As a concrete example, let  $L$  be the unit circle

$$L = \{z \in \mathbb{C} : |z| = 1\},$$

and choose three distinct points on  $L$ :

$$a = 1, \quad b = -1, \quad c = i.$$

Then the associated map

$$\Phi(z) = \frac{(z-a)(c-b)}{(z-b)(c-a)}$$

sends  $L$  onto the real axis, with

$$\Phi(a) = 0, \quad \Phi(b) = \infty, \quad \Phi(c) = 1.$$

For  $z$  not on  $L$ , the sign of  $\Im\Phi(z)$  indicates on which side of the circle  $L$  the point  $z$  lies. One side of  $L$  (for instance, the interior) is mapped to the upper half-plane  $\Im w > 0$ , while the other side (the exterior) is mapped to the lower half-plane  $\Im w < 0$ .

Figure 8 illustrates this: we show the circle  $L$ , the points  $a, b, c$ , and a cloud of sample points in the plane, distinguished according to the sign of  $\Im\Phi(z)$ . In Figure 9, we see their images under  $\Phi$ , which lie strictly above or below the real axis.

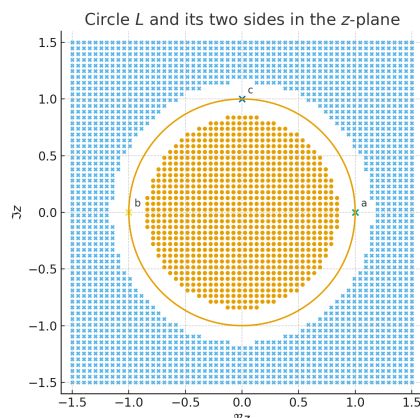


Figure 8: The unit circle  $L$  (example of a generalized line) with three marked points  $a, b, c \in L$  and sample points on its two sides. The two sides are distinguished according to the sign of  $\Im\Phi(z)$ , which will become the upper and lower half-planes in the  $w$ -plane.

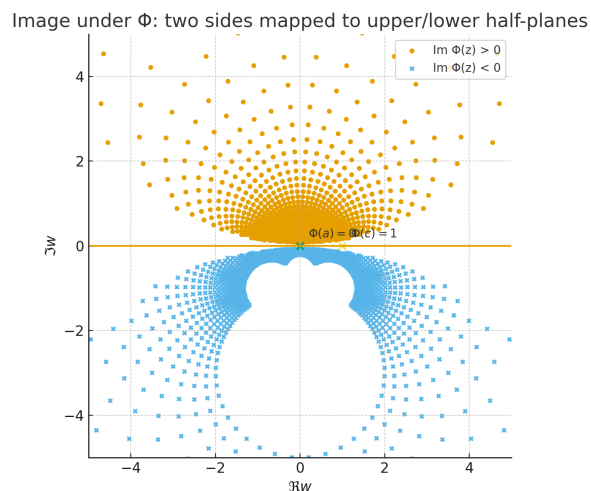


Figure 9: Image of the configuration under the cross-ratio map  $\Phi$ . The circle  $L$  is sent to the real axis, with  $\Phi(a) = 0$ ,  $\Phi(b) = \infty$ ,  $\Phi(c) = 1$ . Points on one side of  $L$  land in the upper half-plane  $\Im w > 0$ , and points on the other side land in the lower half-plane  $\Im w < 0$ .

**Example: a line mapped to the real axis**

Now let  $L$  be a straight line instead of a circle. Take, for instance, the imaginary axis

$$L = \{z = iy : y \in \mathbb{R}\},$$

and choose three distinct points on  $L$ :

$$a = i, \quad b = -i, \quad c = 0.$$

The corresponding cross-ratio map

$$\Phi(z) = \frac{(z - a)(c - b)}{(z - b)(c - a)}$$

again satisfies

$$\Phi(a) = 0, \quad \Phi(b) = \infty, \quad \Phi(c) = 1,$$

and sends the whole line  $L$  onto the real axis  $\mathbb{R} \cup \{\infty\}$ . As before, the two sides of  $L$  are mapped onto the upper and lower half-planes:

$$\text{one side of } L \mapsto \{w : \Im w > 0\}, \quad \text{the other side} \mapsto \{w : \Im w < 0\}.$$

Figure 10 shows the line  $L$  and some sample points on each side, distinguished by the sign of  $\Im \Phi(z)$ . Figure 11 illustrates their images under  $\Phi$ : the line becomes the real axis, with  $\Phi(a) = 0$  and  $\Phi(c) = 1$ , and the two sides become the upper and lower half-planes.

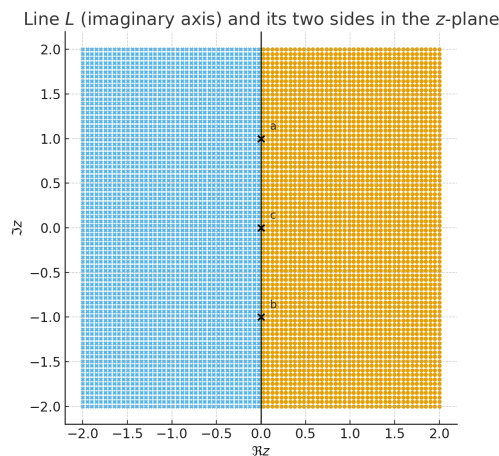


Figure 10: The line  $L$  (the imaginary axis) with marked points  $a = i$ ,  $b = -i$ ,  $c = 0$ . Sample points on its two sides are labelled according to the sign of  $\Im \Phi(z)$ .

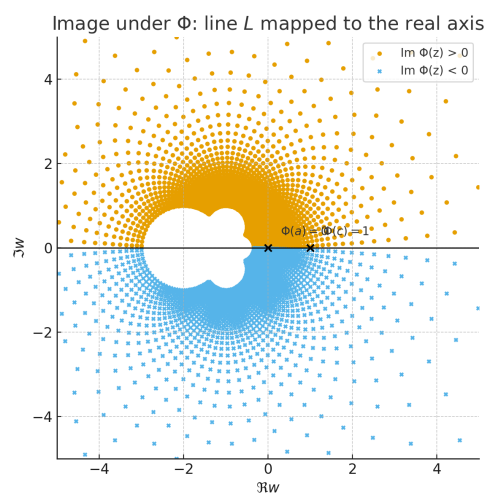


Figure 11: Image of the configuration under  $\Phi$ . The line  $L$  is mapped to the real axis, with  $\Phi(a) = 0$ ,  $\Phi(b) = \infty$ ,  $\Phi(c) = 1$ . One side of  $L$  maps to the upper half-plane  $\Im w > 0$ , the other side to the lower half-plane  $\Im w < 0$ .

**Half-plane to half-plane via three boundary points**

Let  $H_1, H_2$  be half-planes with boundary arcs  $L_1, L_2$ . Choose boundary triples

$$a, b, c \in L_1, \quad A, B, C \in L_2$$

in the same cyclic order as seen from the chosen sides of  $H_1, H_2$ .

First send  $L_1$  to the real axis by the cross-ratio map

$$\Phi(z) := \frac{(z-a)(c-b)}{(z-b)(c-a)}.$$

Then  $\Phi(a) = 0$ ,  $\Phi(b) = \infty$ ,  $\Phi(c) = 1$ , and the two sides of  $L_1$  correspond to the half-planes  $\Im \Phi(z) \gtrless 0$ .

Next send  $L_2$  to the real axis by

$$\Psi(w) := \frac{(w-A)(C-B)}{(w-B)(C-A)},$$

so that  $\Psi(A) = 0$ ,  $\Psi(B) = \infty$ ,  $\Psi(C) = 1$ .

The desired map  $T : H_1 \rightarrow H_2$  is given by

$$\Psi(T(z)) = \Phi(z), \quad \text{i.e.} \quad T(z) = \Psi^{-1}(\Phi(z)).$$

Solving for  $\Psi^{-1}$  yields the explicit formula

$$T(z) = \frac{A(C-B) - \Phi(z)B(C-A)}{(C-B) - \Phi(z)(C-A)}, \quad \Phi(z) = \frac{(z-a)(c-b)}{(z-b)(c-a)}.$$

Then

$$T(a) = A, \quad T(b) = B, \quad T(c) = C,$$

and  $T$  maps the chosen side of  $H_1$  biholomorphically onto the chosen side of  $H_2$ . If the orientation of the half-plane is reversed, one may replace  $\Phi$  by  $-\Phi$  or interchange two of the boundary points to flip the side.

**Example: upper half-plane to right half-plane**

A particularly simple instance is obtained by taking

$$T(z) = -iz.$$

Then

$$\Re T(z) = \Re(-iz) = \Im z,$$

so

$$\Im z > 0 \iff \Re T(z) > 0.$$

Thus  $T$  maps the upper half-plane

$$H_1 = \{z : \Im z > 0\}$$

bijectionally onto the right half-plane

$$H_2 = \{w : \Re w > 0\},$$

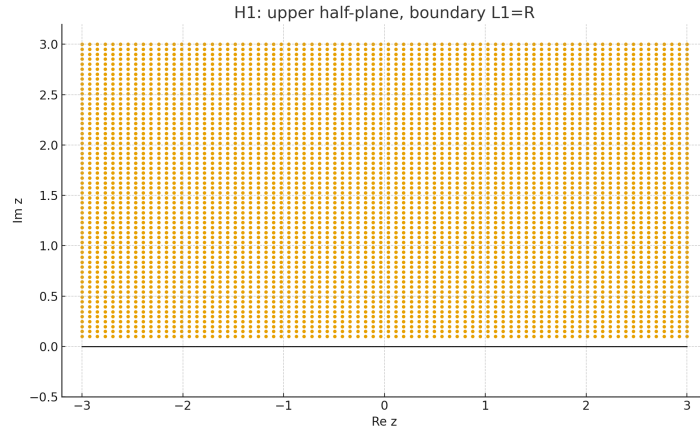


Figure 12: Source half-plane  $H_1 = \{\Im z > 0\}$  with boundary  $L_1 = \mathbb{R}$  and a grid of sample points.

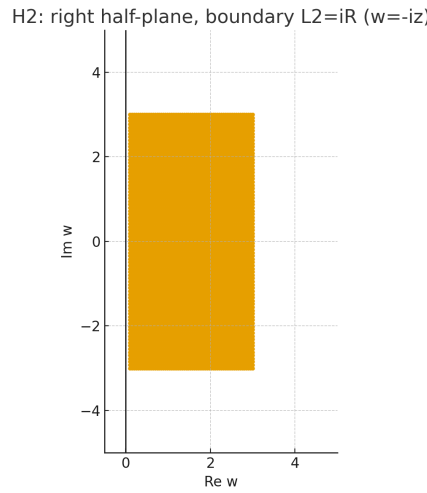


Figure 13: Image under  $T(z) = -iz$ : the upper half-plane is mapped to the right half-plane  $H_2 = \{\Re w > 0\}$ , and the real axis is mapped to the imaginary axis  $L_2 = i\mathbb{R}$ . All sample points lie strictly to the right of the boundary line.

sending the boundary line  $L_1 = \mathbb{R}$  to the boundary line  $L_2 = i\mathbb{R}$ .

Figures 12 and 13 illustrate this example.

The above example is the simplest possible case of a half-plane to half-plane mapping, since  $T(z) = -iz$  is just a rotation by  $-\pi/2$ . However, every biholomorphic map between half-planes can be obtained from the general three-boundary-points construction described earlier. In particular, if  $a, b, c$  are three distinct points on the boundary line  $L_1 = \mathbb{R}$ , and  $A, B, C$  are three distinct points on the boundary line  $L_2 = i\mathbb{R}$  in the same cyclic order, then the unique Möbius map satisfying

$$T(a) = A, \quad T(b) = B, \quad T(c) = C$$

is given explicitly by

$$T(z) = \frac{A(C - B) - \Phi(z) B(C - A)}{(C - B) - \Phi(z) (C - A)}, \quad \Phi(z) = \frac{(z - a)(c - b)}{(z - b)(c - a)}.$$

This expression reduces to  $T(z) = -iz$  precisely when the triples  $(a, b, c)$  and  $(A, B, C)$  are chosen so that the cross-ratios coincide and the real axis is mapped to the imaginary axis by a pure rotation. Thus the simple rotation  $T(z) = -iz$  is not an isolated coincidence, but a special case of the general boundary-normalization principle.

### Non-trivial example: half-plane to half-plane via three boundary points

We now give a non-trivial instance of the general construction. Take three distinct boundary points on the real axis

$$a = -2, \quad b = 1, \quad c = 4,$$

and three distinct boundary points on the imaginary axis

$$A = -i, \quad B = 0, \quad C = 2i,$$

listed in the same cyclic order (bottom  $\rightarrow$  top).

The cross-ratio map sending  $a, b, c$  to  $0, \infty, 1$  is

$$\Phi(z) = \frac{(z-a)(c-b)}{(z-b)(c-a)} = \frac{(z+2)(3)}{(z-1)(6)}.$$

The inverse map sending  $0, \infty, 1$  back to  $A, B, C$  is

$$\Psi^{-1}(\xi) = \frac{A(C-B) - \xi B(C-A)}{(C-B) - \xi(C-A)}.$$

Hence the desired half-plane map is

$$T(z) = \Psi^{-1}(\Phi(z)),$$

which explicitly is

$$T(z) = \frac{-i(2i-0) - \Phi(z)0(2i-(-i))}{(2i-0) - \Phi(z)(2i-(-i))} = \frac{-2i^2}{2i-3i\Phi(z)} = \frac{2}{2i-3i\Phi(z)}.$$

With these boundary triples one verifies

$$T(a) = A, \quad T(b) = B, \quad T(c) = C,$$

and  $T$  maps the chosen side  $\Im z > 0$  biholomorphically onto the chosen side  $\Re w > 0$ .

Figures 14 and 15 illustrate this map.

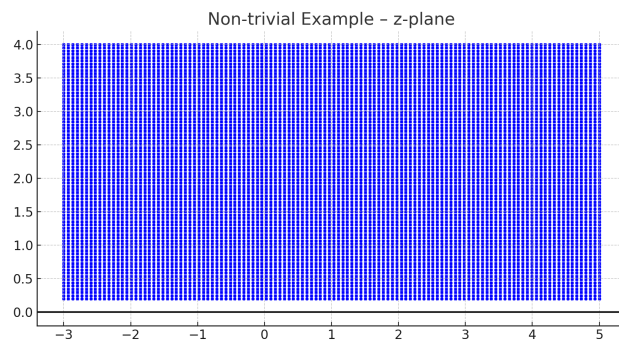


Figure 14: Non-trivial boundary triple on  $L_1 = \mathbb{R}$ :  $a = -2$ ,  $b = 1$ ,  $c = 4$ . A grid of sample points in the upper half-plane is shown.

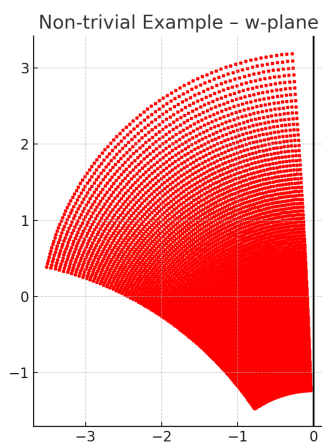


Figure 15: Image under the non-trivial map  $T(z) = \Psi^{-1}(\Phi(z))$ . The real axis is sent to the imaginary axis with  $T(a) = A = -i$ ,  $T(b) = B = 0$ ,  $T(c) = C = 2i$ , and every interior point satisfies  $\Re T(z) > 0$ .

## Half-plane to half-plane via three boundary points

Let  $H_1, H_2$  be half-planes with boundary arcs  $L_1, L_2$ . Choose boundary triples

$$a, b, c \in L_1, \quad A, B, C \in L_2$$

in the same cyclic order as seen from the chosen sides of  $H_1, H_2$ .

First send  $L_1$  to the real axis by the cross-ratio map

$$\Phi(z) := \frac{(z-a)(c-b)}{(z-b)(c-a)}.$$

Then

$$\Phi(a) = 0, \quad \Phi(b) = \infty, \quad \Phi(c) = 1,$$

and the two sides of  $L_1$  correspond to the half-planes  $\Im \Phi(z) \gtrless 0$ .

Next send  $L_2$  to the real axis by

$$\Psi(w) := \frac{(w-A)(C-B)}{(w-B)(C-A)},$$

so that

$$\Psi(A) = 0, \quad \Psi(B) = \infty, \quad \Psi(C) = 1.$$

The desired half-plane map  $T : H_1 \rightarrow H_2$  is obtained by requiring

$$\Psi(T(z)) = \Phi(z),$$

that is,

$$T(z) = \Psi^{-1}(\Phi(z)).$$

Solving explicitly for  $\Psi^{-1}$  gives

$$T(z) = \frac{A(C-B) - \Phi(z)B(C-A)}{(C-B) - \Phi(z)(C-A)}, \quad \Phi(z) = \frac{(z-a)(c-b)}{(z-b)(c-a)}.$$

Then

$$T(a) = A, \quad T(b) = B, \quad T(c) = C,$$

and  $T$  maps the chosen side of  $H_1$  biholomorphically onto the chosen side of  $H_2$ .

### Example: automorphism of the upper half-plane

Consider the Möbius transformation

$$T(z) = \frac{z-1}{z+1}.$$

We work on the upper half-plane

$$H = \{z : \Im z > 0\}$$

with boundary the extended real line

$$L = \mathbb{R} \cup \{\infty\}.$$

**Preservation of the half-plane.** Writing  $z = x + iy$  with  $y > 0$ , a direct computation yields

$$\Im T(z) = \Im \left( \frac{z-1}{z+1} \right) = \frac{\Im z}{|z+1|^2} = \frac{y}{|z+1|^2} > 0.$$

Hence

$$\Im z > 0 \iff \Im T(z) > 0,$$

so  $T$  is a non-affine automorphism of the upper half-plane and maps  $L$  to itself.

**Boundary triples.** Along the boundary  $L$  we have:

$$T(1) = \frac{1-1}{1+1} = 0, \quad T(-1) = \frac{-1-1}{-1+1} = \infty, \quad T(\infty) = \lim_{z \rightarrow \infty} \frac{z-1}{z+1} = 1.$$

Thus, in the three-point normalization language we may choose

$$a = 1, \quad b = -1, \quad c = \infty \quad \text{on } L_1 = L,$$

and

$$A = 0, \quad B = \infty, \quad C = 1 \quad \text{on } L_2 = L.$$

Then  $T$  is exactly the Möbius map sending

$$a \mapsto A, \quad b \mapsto B, \quad c \mapsto C,$$

i.e.

$$1 \mapsto 0, \quad -1 \mapsto \infty, \quad \infty \mapsto 1.$$

When one of the boundary points is  $\infty$ , the cross-ratio formula simplifies: for  $(a, b, c) = (1, -1, \infty)$  we have

$$\Phi(z) = \frac{(z-a)(c-b)}{(z-b)(c-a)} = \frac{(z-1)}{(z+1)},$$

so in this case  $T(z) = \Phi(z)$  is precisely the three-point normalization carrying  $(1, -1, \infty)$  to  $(0, \infty, 1)$  on the same boundary line.

Figures 16 and 17 illustrate the action of  $T$  on the upper half-plane.

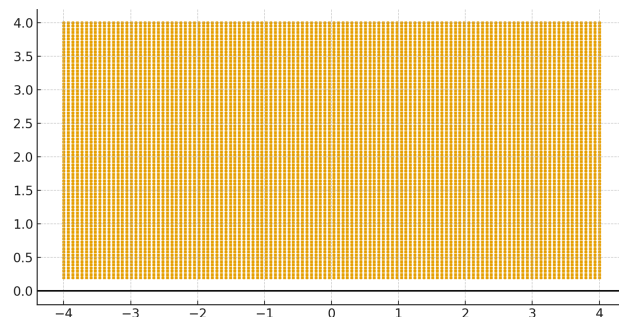


Figure 16: Upper half-plane  $H = \{\Im z > 0\}$  with a grid of sample points. The boundary is the real axis  $L = \mathbb{R}$ .

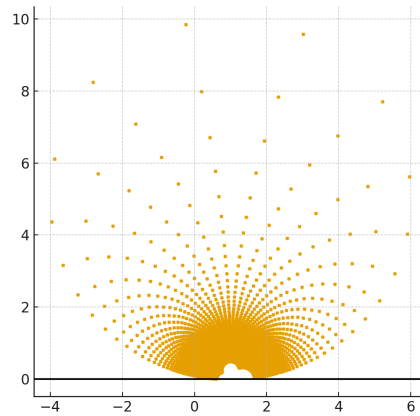


Figure 17: Image of the grid under  $T(z) = \frac{z-1}{z+1}$ . The real axis remains the boundary, and the interior of the upper half-plane is mapped to itself in a non-trivial way.

### Half-plane to half-plane via three boundary points (non-automorphism)

Let

$$H_1 = \{z : \Im z > 0\}, \quad L_1 = \mathbb{R},$$

and

$$H_2 = \{w : \Re w > 1\}, \quad L_2 = \{w : \Re w = 1\}.$$

Choose boundary triples

$$a = -1, \quad b = 0, \quad c = 2 \in L_1,$$

$$A = 1 - i, \quad B = 1, \quad C = 1 + i \in L_2,$$

listed in the same cyclic order (bottom to top).

The cross-ratio map sending  $L_1$  to the real axis is

$$\Phi(z) = \frac{(z-a)(c-b)}{(z-b)(c-a)} = \frac{2(z+1)}{3z},$$

so that  $\Phi(a) = 0$ ,  $\Phi(b) = \infty$ ,  $\Phi(c) = 1$ . Similarly, the map sending  $L_2$  to the real axis is

$$\Psi(w) = \frac{(w-A)(C-B)}{(w-B)(C-A)} = \frac{w-1+i}{2(w-1)},$$

for which  $\Psi(A) = 0$ ,  $\Psi(B) = \infty$ ,  $\Psi(C) = 1$ .

The desired map  $T : H_1 \rightarrow H_2$  is defined by

$$\Psi(T(z)) = \Phi(z),$$

i.e.  $T(z) = \Psi^{-1}(\Phi(z))$ . Solving  $\Psi(w) = \xi$  for  $w$  gives

$$\Psi^{-1}(\xi) = \frac{2\xi - 1 + i}{2\xi - 1},$$

and hence

$$T(z) = \frac{2\Phi(z) - 1 + i}{2\Phi(z) - 1}, \quad \Phi(z) = \frac{2(z+1)}{3z}.$$

Then

$$T(-1) = 1 - i, \quad T(0) = 1, \quad T(2) = 1 + i,$$

and  $T$  maps the upper half-plane  $\Im z > 0$  biholomorphically onto the half-plane  $\Re w > 1$ .

Figures 18 and 19 illustrate this non-automorphism half-plane map.

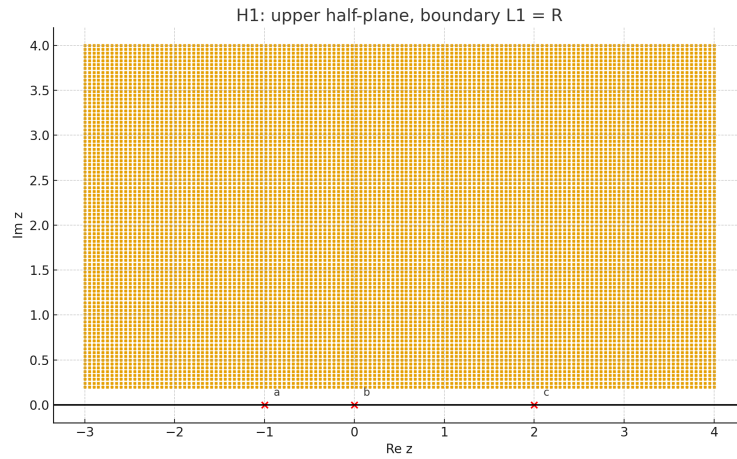


Figure 18: Source half-plane  $H_1 = \{\Im z > 0\}$ , boundary  $L_1 = \mathbb{R}$ , and boundary points  $a = -1$ ,  $b = 0$ ,  $c = 2$ .

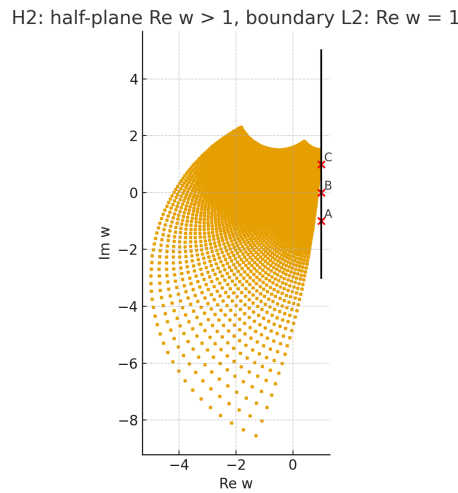


Figure 19: Image under  $T(z) = \Psi^{-1}(\Phi(z))$ . The upper half-plane is mapped to the shifted right half-plane  $H_2 = \{\Re w > 1\}$ , and the boundary points satisfy  $T(a) = 1 - i$ ,  $T(b) = 1$ ,  $T(c) = 1 + i$ .

## Disk automorphisms

Let

$$\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$$

denote the unit disk. For  $a \in \mathbb{D}$  and  $\theta \in \mathbb{R}$  define

$$\phi_{a,\theta}(z) := e^{i\theta} \frac{z - a}{1 - \bar{a}z}, \quad |a| < 1.$$

The map  $\phi_{a,\theta}$  is a Möbius transformation which maps  $\mathbb{D}$  biholomorphically onto itself. Indeed, the basic factor

$$\psi_a(z) := \frac{z - a}{1 - \bar{a}z}$$

satisfies  $\psi_a(a) = 0$  and  $\psi_a(0) = -a$ . A direct computation gives

$$|\psi_a(z)|^2 = 1 - \frac{(1 - |a|^2)(1 - |z|^2)}{|1 - \bar{a}z|^2},$$

so in particular  $|\psi_a(z)| < 1$  whenever  $|z| < 1$ , and  $|\psi_a(z)| = 1$  when  $|z| = 1$ . Multiplication by the unimodular factor  $e^{i\theta}$  is a rotation of the disk, hence

$$\phi_{a,\theta}(\mathbb{D}) = \mathbb{D}, \quad \phi_{a,\theta}(\partial\mathbb{D}) = \partial\mathbb{D}.$$

By the Schwarz–Pick theorem, every holomorphic automorphism of  $\mathbb{D}$  is of this form.

### Example

Take

$$a = 0.5 + 0.2i, \quad \theta = \frac{\pi}{4},$$

and define

$$\phi(z) = \phi_{a,\theta}(z) = e^{i\pi/4} \frac{z - a}{1 - \bar{a}z}.$$

Then  $\phi(a) = 0$ , and  $\phi$  maps  $\mathbb{D}$  onto itself while distorting the radial/circular grid in a non-trivial way.

Figures 20 and 21 show the unit disk and its image under  $\phi$ .

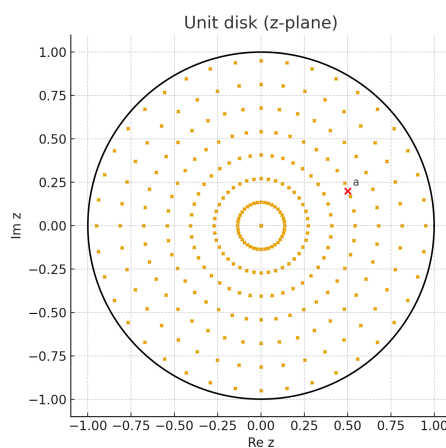


Figure 20: Unit disk  $\mathbb{D}$  with a polar grid and the point  $a = 0.5 + 0.2i$  marked.

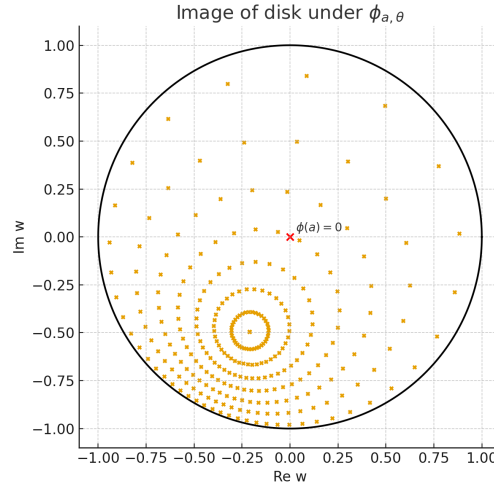


Figure 21: Image of the grid under  $\phi_{a,\theta}(z) = e^{i\theta} \frac{z-a}{1-\bar{a}z}$  with  $a = 0.5 + 0.2i$  and  $\theta = \pi/4$ . The unit circle is preserved and  $\phi(a) = 0$ .

## Strip to right half-plane: extended geometry

Consider the strip

$$S_h = \{z = x + iy : |\Im z| < h/2\}.$$

The exponential map

$$w = \exp\left(\frac{\pi z}{h}\right)$$

sends  $S_h$  conformally onto the right half-plane  $\Re w > 0$ .

Writing  $z = x + iy$ , we obtain the polar decomposition

$$w = e^{\pi x/h} e^{i\pi y/h} = r e^{i\theta}, \quad r = e^{\pi x/h}, \quad \theta = \frac{\pi y}{h}.$$

**Boundary mapping.** The boundary lines  $y = \pm h/2$  satisfy

$$\theta = \pm\pi/2,$$

so the entire boundary of the strip is mapped to the imaginary axis  $\Re w = 0$ . Hence  $S_h$  is mapped onto the right half-plane.

**Images of characteristic curves.**

- **Horizontal lines**  $y = y_0$  have constant argument  $\theta = \pi y_0/h$  and map to *rays from the origin*.
- **Vertical lines**  $x = x_0$  have constant modulus  $r = e^{\pi x_0/h}$  and map to *circles centered at the origin*.
- **Slanted lines**  $y = mx + c$  satisfy  $\theta = m \ln r + \text{const}$  and therefore map to *logarithmic spirals*.
- A rectangular grid in  $S_h$  becomes a *polar grid* of rays and circles in the right half-plane.

## Strip to right half-plane: extended geometry

Consider the strip

$$S_h = \{z = x + iy : |\Im z| < h/2\}.$$

The exponential map

$$w = \exp\left(\frac{\pi z}{h}\right)$$

sends  $S_h$  conformally onto the right half-plane  $\Re w > 0$ .

Writing  $z = x + iy$ , we obtain the polar decomposition

$$w = e^{\pi x/h} e^{i\pi y/h} = r e^{i\theta}, \quad r = e^{\pi x/h}, \quad \theta = \frac{\pi y}{h}.$$

**Boundary mapping.** The boundary lines  $y = \pm h/2$  satisfy

$$\theta = \pm \frac{\pi}{2},$$

so the entire boundary of the strip is mapped to the imaginary axis  $\Re w = 0$ . Hence  $S_h$  is mapped onto the right half-plane.

### Images of characteristic curves (A–E).

(A) **Horizontal lines**  $y = y_0$ .

Here  $\theta = \frac{\pi y_0}{h}$  is constant while  $r = e^{\pi x/h}$  varies. Thus each horizontal line is mapped to a ray

$$\{w : \arg w = \frac{\pi y_0}{h}, \quad r > 0\}$$

emanating from the origin.

(B) **Vertical lines**  $x = x_0$ .

Here  $r = e^{\pi x_0/h}$  is constant while  $\theta$  varies. Hence each vertical line is mapped to a circle

$$\{w : |w| = e^{\pi x_0/h}\}$$

centered at the origin.

(C) **Slanted lines**  $y = mx + c$ .

Then

$$\theta = \frac{\pi}{h}(mx + c), \quad r = e^{\pi x/h}.$$

Eliminating  $x$  gives

$$\theta = m \log r + \text{const},$$

i.e. a *logarithmic spiral*  $r = Ae^{\theta/m}$  for some  $A > 0$ .

(D) **General real line**  $ax + by = c$ .

Writing this in the form  $y = mx + c'$  (when  $b \neq 0$ ) reduces to case (C). If  $b = 0$  we are in case (B). Thus any straight line in the strip is mapped either to a circle, a ray, or a logarithmic spiral in the  $w$ -plane.

(E) **Rectangular grids.**

A rectangular grid of horizontal and vertical lines in  $S_h$  becomes a *polar grid* in the right half-plane: horizontal lines give rays (case (A)), vertical lines give circles (case (B)).

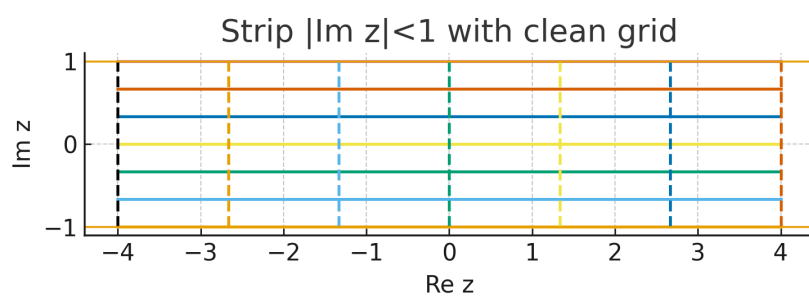


Figure 22: Strip  $|\Im z| < 1$  ( $h = 2$ ) with a clean rectangular grid: horizontal (solid) and vertical (dashed) lines.

Image: rays (solid) & circles (dashed)

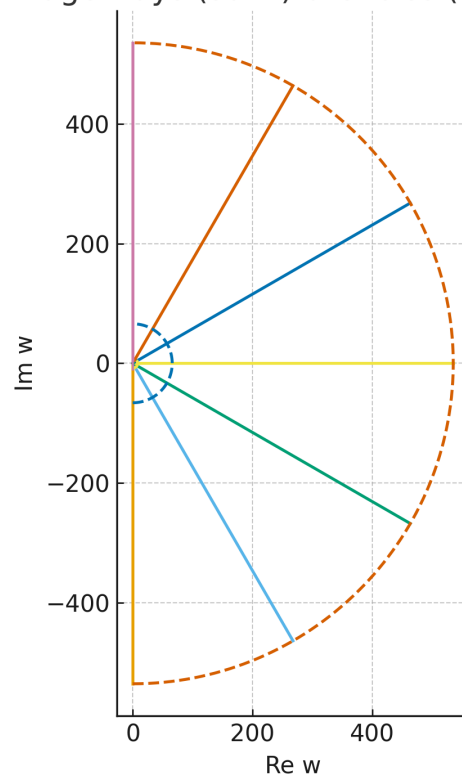


Figure 23: Image under  $w = e^{\pi z/2}$ : horizontal lines become rays, vertical lines become circles centered at the origin.

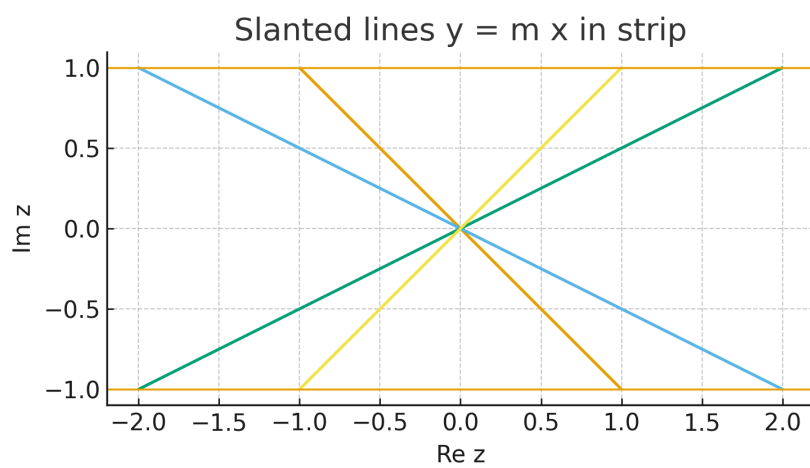


Figure 24: Slanted lines  $y = mx$  intersected with the strip  $|\Im z| < 1$ .

Images of slanted lines: logarithmic spirals

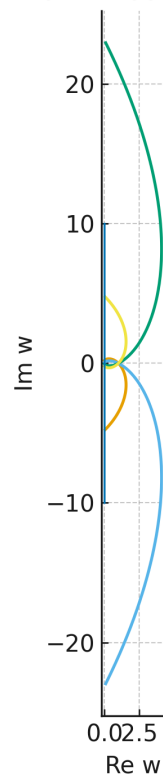


Figure 25: Images of the slanted lines under  $w = e^{\pi z/2}$ . Each straight line in the strip becomes a logarithmic spiral in the right half-plane.